

AI / Machine Learning Roadmap (Beginner to Job Ready)

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Welcome to my AI learning journey!

This document contains my structured roadmap to become job-ready in Artificial Intelligence, Machine Learning, and Data Science.

My Goal

- Become skilled in AI / ML / Data Science
- Build real-world projects
- Create a strong GitHub portfolio
- Get internship + job opportunities

Roadmap Overview

Phase 1: Python Programming Foundation (Week 1–4)

Topics to Learn

- Variables & Data Types
- If-Else Conditions
- Loops (for, while)
- Functions
- Lists, Tuples, Sets, Dictionaries
- Strings
- File Handling
- OOP Concepts (Class, Object, Inheritance)
- Exception Handling

Practice Tasks

- 50+ Python programs
- Pattern problems
- Small mini projects (calculator, quiz app)

Phase 2: Mathematics for AI (Week 5–6)

Topics

- Linear Algebra (Vectors, Matrices)
- Probability Basics
- Statistics (Mean, Median, Mode)

- Variance & Standard Deviation
- Correlation & Regression basics

Phase 3: Data Analysis & Visualization (Week 7–9)

Libraries to Learn

- NumPy
- Pandas
- Matplotlib
- Seaborn

Skills

- Data Cleaning
- Handling Missing Values
- Removing Duplicates
- Data Visualization
- Feature Understanding

Phase 4: Machine Learning (Week 10–14)

Concepts

- Supervised Learning
- Unsupervised Learning
- Train-Test Split
- Overfitting & Underfitting
- Cross Validation
- Feature Scaling
- Model Evaluation Metrics

Algorithms to Learn

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest
- KNN
- Naive Bayes
- Support Vector Machine (SVM)
- K-Means Clustering

Tools

- Scikit-learn

Phase 5: Deep Learning (Week 15–18)

Topics

- Neural Networks basics
- Activation Functions (ReLU, Sigmoid, Softmax)
- Backpropagation
- Loss Functions
- Optimizers (SGD, Adam)
- CNN basics

Libraries

- TensorFlow + Keras OR PyTorch

Project

- Digit recognition using MNIST dataset

Phase 6: NLP (Natural Language Processing) (Week 19–21)

Topics

- Tokenization
- Stopwords removal
- Stemming & Lemmatization
- TF-IDF
- Word Embeddings (Word2Vec)
- Sentiment Analysis
- Text Classification

Projects

- Sentiment analysis on movie reviews
- Spam message classifier

Phase 7: Computer Vision (Week 22–24)

Topics

- OpenCV Basics
- Image Processing

- CNN for image classification
- Object detection basics

Projects

- Face detection system
- Image classification model

Phase 8: Real-World Projects (Week 25–28)

Portfolio Projects (Must)

- House Price Prediction
- Customer Churn Prediction
- Fake News Detection
- Recommendation System
- Sales Prediction
- Loan Approval Prediction

Phase 9: Deployment (Week 29–32)

Skills to Learn

- Model saving (Pickle / Joblib)
- Streamlit
- Flask / FastAPI

Deployment platforms

- Render
- Railway
- Hugging Face Spaces

Deployment Project

- Deploy ML model as a web app using Streamlit

Phase 10: Interview Preparation (Week 33–36)

Topics

- Python interview questions
- ML interview questions
- SQL basics
- DSA basics (arrays, strings, recursion)
- Resume + LinkedIn optimization

Tools & Platforms to Learn

- Git & GitHub
- Kaggle
- Google Colab
- Jupyter Notebook
- VS Code
- ChatGPT (for learning & debugging)

Recommended Daily Routine

- 1 hour Learning
- 1 hour Coding practice
- 30 min revision
- 30 min project work

Progress Tracker

- Python Basics
- NumPy
- Pandas
- Data Visualization
- Machine Learning
- Deep Learning
- NLP
- Computer Vision
- Deployment
- Portfolio Projects
- Interview Preparation

Final Target

- Strong AI/ML skills
- 5+ projects on GitHub
- Internship-ready portfolio
- Job-ready AI Engineer